

Pre-Workshop Task (C Language)

Task: "Student Record Mini-Program"

Write a C program to:

1. Accept a student's **roll number, name, age, and marks** (in 3 subjects) using appropriate datatypes
2. Calculate the **total and percentage**
3. Using **if-else**, print the **grade** based on percentage:
 - o ≥ 90 : A+, ≥ 75 : A, ≥ 60 : B, ≥ 40 : C, else $\rightarrow$ Fail
4. Use a **loop** to accept and process this for **5 students** (not just 1)
5. At the end, print the name of the student with the **highest percentage**

Task 2: "Simple Calculator with Menu"

Write a C program to:

1. Display a menu: Addition, Subtraction, Multiplication, Division, Modulus
2. Accept two numbers and the user's choice using a **switch-case**
3. Perform the selected operation and display the result
4. Handle **division by zero** with an appropriate error message
5. Use a **loop** to let the user perform multiple calculations until they choose "Exit"

Task 3: "Basic Statistics Calculator"

Write a C program to:

1. Accept **N numbers** (let the user decide how many) into an array
2. Calculate and display:
 - o **Sum and Average**
 - o **Maximum and Minimum** value
 - o **Count of even and odd numbers**
3. Calculate how many numbers are **above average**

Task 4: "Number Properties Checker"

Write a C program to:

1. Accept a number from the user
2. Check and display whether it is:
 - o **Even or Odd**
 - o **Prime or Not Prime**
 - o **Palindrome or Not** (e.g., 121, 1331)
 - o **Sum of digits**
3. Use a **menu-driven loop** so the user can check multiple numbers without restarting the program

Task 5: "Simple Banking Simulator (Procedural, no OOP)"

Write a C program to:

1. Store a starting balance for **one account** (just a variable, no struct required)
2. Provide a menu: Deposit, Withdraw, Check Balance, Exit
3. Validate: withdrawal cannot exceed current balance; deposit/withdrawal cannot be negative
4. Loop the menu until the user exits